

CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

# Chapter 7: Coordinate Geometry

Topic-wise short notes with solved examples

With coordinates, lengths and positions can be worked out by calculation. This chapter has two tools: the distance formula and the section formula.

## 1 Coordinates: a quick recap

- A point in the plane is written  $(x, y)$ .  $x$  is the **abscissa** (distance from the  $y$ -axis) and  $y$  is the **ordinate** (distance from the  $x$ -axis).
- A point on the  $x$ -axis has the form  $(x, 0)$ . A point on the  $y$ -axis has the form  $(0, y)$ . The origin is  $(0, 0)$ .
- Signs in the quadrants: I  $(+, +)$ , II  $(-, +)$ , III  $(-, -)$ , IV  $(+, -)$ .

## 2 The distance formula

- The distance between  $P(x_1, y_1)$  and  $Q(x_2, y_2)$  is

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- The distance of  $P(x, y)$  from the origin is  $\sqrt{x^2 + y^2}$ .
- The formula comes from the Pythagoras theorem: the horizontal gap is  $x_2 - x_1$  and the vertical gap is  $y_2 - y_1$ .

**Example 2.1** Find the distance between  $A(3, -4)$  and  $B(-1, -1)$ .

**SOLUTION**

$$AB = \sqrt{(-1 - 3)^2 + (-1 - (-4))^2} = \sqrt{(-4)^2 + 3^2} = \sqrt{16 + 9}$$

$$AB = 5 \text{ units}$$

**Example 2.2** Find the values of  $k$  if the distance between  $A(3, k)$  and  $B(-1, 2)$  is 5 units.

**SOLUTION**

$$\sqrt{(-1 - 3)^2 + (2 - k)^2} = 5, \text{ so } 16 + (2 - k)^2 = 25$$

$$(2 - k)^2 = 9, \text{ so } 2 - k = 3 \text{ or } 2 - k = -3$$

$$k = -1 \text{ or } k = 5$$

**Example 2.3** Find the point on the  $x$ -axis that is equidistant from  $A(3, 4)$  and  $B(7, 6)$ .

**SOLUTION**

Let the point be  $P(x, 0)$ . Then  $PA^2 = PB^2$ :  $(x - 3)^2 + 16 = (x - 7)^2 + 36$

$x^2 - 6x + 25 = x^2 - 14x + 85$ , so  $8x = 60$  and  $x = 7.5$

**The point is  $(7.5, 0)$ .**

**REMEMBER**

When two distances are equal, equate their **squares**. This removes the square roots at once.

### 3 Using distances to identify figures

- **Collinear points:** three points  $A, B, C$  lie on one line if the sum of two of the distances  $AB, BC, CA$  equals the third.
- **Triangle:** isosceles if two sides are equal; equilateral if all three are equal; right-angled if the squares of two sides add up to the square of the third.

| Quadrilateral | What to show                                         |
|---------------|------------------------------------------------------|
| Parallelogram | Opposite sides are equal                             |
| Rectangle     | Opposite sides are equal and the diagonals are equal |
| Rhombus       | All four sides are equal                             |
| Square        | All four sides are equal and the diagonals are equal |

**Example 3.1** Show that the points  $A(1, 1)$ ,  $B(4, 5)$  and  $C(7, 9)$  are collinear.

**SOLUTION**

$AB = \sqrt{9 + 16} = 5$ ,  $BC = \sqrt{9 + 16} = 5$ ,  $AC = \sqrt{36 + 64} = 10$

$AB + BC = 5 + 5 = 10 = AC$

**The points are collinear.**

**Example 3.2** Show that  $A(1, 2)$ ,  $B(4, 6)$  and  $C(8, 3)$  are the vertices of an isosceles right-angled triangle.

**SOLUTION**

$AB^2 = 9 + 16 = 25$ ,  $BC^2 = 16 + 9 = 25$ ,  $AC^2 = 49 + 1 = 50$

$AB = BC$ , so the triangle is isosceles.

$AB^2 + BC^2 = 50 = AC^2$ , so by the converse of the Pythagoras theorem the angle at  $B$  is  $90^\circ$ .

**$\triangle ABC$  is an isosceles right-angled triangle, right-angled at  $B$ .**

## 4 The section formula

- The point  $P$  that divides the line segment joining  $A(x_1, y_1)$  and  $B(x_2, y_2)$  **internally** in the ratio  $m_1 : m_2$  is

$$P = \left( \frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)$$

- $m_1$  goes with the **far** point:  $m_1$  multiplies  $(x_2, y_2)$  and  $m_2$  multiplies  $(x_1, y_1)$ .
- Mid-point** (ratio 1 : 1):

$$M = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- To **find the ratio** in which a point divides a segment, take the ratio as  $k : 1$ .

**Example 4.1** Find the point that divides the segment joining  $A(1, -2)$  and  $B(6, 8)$  in the ratio 2 : 3.

**SOLUTION**

$$x = \frac{2 \times 6 + 3 \times 1}{2 + 3} = \frac{15}{5} = 3$$

$$y = \frac{2 \times 8 + 3 \times (-2)}{2 + 3} = \frac{10}{5} = 2$$

**The point is (3, 2).**

**Example 4.2** In what ratio does the  $x$ -axis divide the segment joining  $A(2, -3)$  and  $B(5, 6)$ ? Find the point of division.

**SOLUTION**

Let the ratio be  $k : 1$ . On the  $x$ -axis the  $y$ -coordinate is 0:  $\frac{6k - 3}{k + 1} = 0$ , so  $k = \frac{1}{2}$ .

The ratio is 1 : 2. Then  $x = \frac{1 \times 5 + 2 \times 2}{1 + 2} = 3$ .

**Ratio 1 : 2; the point is (3, 0).**

**Example 4.3** Find the points of trisection of the segment joining  $A(1, 2)$  and  $B(7, 11)$ .

**SOLUTION**

The points of trisection divide  $AB$  in the ratios 1 : 2 and 2 : 1.

$$P = \left( \frac{7 + 2}{3}, \frac{11 + 4}{3} \right) = (3, 5) \text{ and } Q = \left( \frac{14 + 1}{3}, \frac{22 + 2}{3} \right) = (5, 8)$$

**The points are (3, 5) and (5, 8).**

**Example 4.4** Three vertices of a parallelogram  $ABCD$ , taken in order, are  $A(1, 2)$ ,  $B(4, 3)$  and  $C(6, 6)$ . Find  $D$ .

**SOLUTION**

The diagonals of a parallelogram bisect each other, so  $AC$  and  $BD$  have the same mid-point.

Mid-point of  $AC = \left(\frac{7}{2}, 4\right)$ . Let  $D = (x, y)$ :  $\frac{4+x}{2} = \frac{7}{2}$  and  $\frac{3+y}{2} = 4$ .

$D = (3, 5)$

## 5 Common mistakes

- Subtracting in different orders for  $x$  and  $y$ . Use  $x_2 - x_1$  and  $y_2 - y_1$ , with the same point first each time.
- Sign errors with negative coordinates:  $-1 - (-4) = 3$ , not  $-5$ .
- Pairing  $m_1$  with  $(x_1, y_1)$  in the section formula.
- Forgetting to write "units" after a distance.
- Claiming a square after showing only four equal sides. That shows a rhombus; the diagonals must also be equal.